



Hourly Gig Economy Platform — Project Specification

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A Limber-style gig platform connecting workers (students & professionals) with employers for hourly-rate shifts. This document covers the complete product feature set, onboarding flows, data model schema, and REST API structure.

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1. Platform Overview

GigWork is a two-sided marketplace where **workers** (students, freelancers, part-time professionals) can discover and apply for hourly shifts posted by **employers** (businesses, event organisers, retailers, etc.). The platform handles the full shift lifecycle: discovery, application, confirmation, geofenced attendance, timesheet, payment, and mutual ratings.

Attribute	Detail
Platform type	Two-sided gig marketplace (B2B2C)
Business model	Commission / service fee on each completed shift
Primary users	Workers (students, part-timers) & Employers (SMEs, enterprises)
Core loop	Post shift → Apply → Confirm → Clock-in/out → Pay → Rate
Auth method	OTP (phone or email)
Payment model	Platform collects from employer; net paid to worker after fee

2. Feature Listing — Worker Side

Authentication

- OTP-based signup via phone or email
- Secure token-based session management (JWT + refresh tokens)
- Biometric re-auth for sensitive actions (payouts)

Profile & Onboarding

- Professional profile with photo, display name, and bio (500 chars)
- Date of birth, location, and preferred work-radius setting
- Field of interest selection (multi-select categories)
- Qualification form: education, work experience, certifications
- Document upload: government ID, right-to-work, background check consent
- Payout account setup (bank account / UPI)

Shift Discovery

- Map and list view of available shifts near worker's location
- Filter by category, date, hourly rate, distance, duration
- Saved search preferences and favourite categories
- Push notifications for new matching shifts

Applications & Scheduling

- One-tap shift application
- Application status tracking (applied → shortlisted → confirmed)
- Upcoming shifts calendar view
- Shift withdrawal before confirmation deadline

On-Shift Tools

- Geofenced clock-in (must be within defined radius of venue)

- Clock-out with automatic hours calculation
- In-app messaging with employer contact during shift

Earnings & Payments

- Earnings dashboard with per-shift breakdown
- Payout history and upcoming scheduled payouts
- Tax summary export (annual)

Ratings & Reputation

- Receive ratings from employers after each shift
 - Rate employers (1–5 stars + comment)
 - Cumulative rating shown on profile
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3. Feature Listing — Employer Side

Business Profile

- Company profile with logo, description, and industry type
- Registration number and address verification
- Multiple admin users per business account
- Billing setup (card or bank account)

Shift Management

- Create shifts: title, description, category, location, date/time, hourly rate, slot count
- Edit or cancel shifts before fill (with auto-notify to applicants)
- Duplicate previous shifts for recurring needs
- Shift status tracking: draft → open → filled → completed

Worker Management

- Browse & filter worker profiles by rating, category, proximity
- Invite specific workers to shifts directly
- View applicant list with profile, qualifications, and rating
- Accept or reject applicants; shortlist multiple candidates
- Favourite worker list for re-hiring

On-Shift Day

- Real-time worker location on shift-day map view
- Clock-in confirmation alerts
- In-app messaging with confirmed workers

Timesheets & Payments

- Timesheet review with logged hours per worker
- Approve, edit, or dispute timesheets
- Automatic payment trigger on approval
- Invoice generation per shift and monthly billing summary

Analytics

- Shift fill rate and average time-to-fill

- Total spend by category, month, or location
 - Average worker rating received
 - Headcount trends over time
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4. Feature Listing — Admin / Platform

- User moderation: suspend, ban, or flag accounts
- KYC management: review & approve/reject uploaded documents
- Dispute resolution centre (worker vs employer timesheet disputes)
- Commission & fee configuration (per category or global)
- Fraud detection flags (unusual clock-in locations, duplicate IDs)
- Platform-wide analytics and revenue reporting
- Category and tag management
- Broadcast notifications to user segments

5. Onboarding Flow — Worker (7 Steps)

Workers complete a structured 7-step onboarding. Steps 1–4 are required to browse shifts. Steps 5–7 are required before a worker can be confirmed for and paid from a shift.

Step 1	OTP Verification
	User enters phone number or email. A 6-digit OTP is sent and must be verified within 5 minutes. On success, a JWT + refresh token pair is issued. • Fields: phone / email • OTP expiry: 5 minutes • Max attempts: 5 before cooldown
Step 2	Basic Info
	Collect the worker's legal name, date of birth, and profile photo. • Fields: full_name, dob, avatar (upload) • DOB used for age verification (18+ required)
Step 3	Location & Radius
	Worker sets their base location and maximum willingness to travel for shifts. • Fields: city, lat/long (device GPS), radius_km • Default radius: 10 km
Step 4	Field of Interest
	Worker selects one or more categories they are willing to work in. • Multi-select from platform category list • Examples: Hospitality, Retail, Logistics, Events, Healthcare Support
Step 5	Qualifications
	Worker fills in their educational background, work experience, and any relevant certifications. • Types: education, work_experience, certification • Fields: title, institution, from_date, to_date, description
Step 6	Document Upload
	Worker uploads identity and eligibility documents for KYC review. • Required: government-issued photo ID, right-to-work document • Optional: background check consent form • Documents reviewed asynchronously by admin within 24–48 hrs

Step 7	Payout Setup
	Worker adds their bank account or UPI ID so platform can transfer earnings after shift approval.
	• Fields: account holder name, account number, IFSC / UPI ID • Verified via small test deposit (optional)

6. Onboarding Flow — Employer (5 Steps)

Employers complete a 5-step onboarding. Steps 1–3 enable shift posting. Step 5 (payment) is required before workers can be paid.

Step 1	OTP Verification
	Business email OTP verification. Creates employer-role user account.
	• Fields: business email • OTP expiry: 10 minutes

Step 2	Company Information
	Core business details for profile and compliance.
	• Fields: company_name, registration_number, industry_type, logo • Registration number used for verification

Step 3	Address & Service Area
	Primary business address and the geographic area from which workers will be drawn.
	• Fields: address, city, lat/lng, service_radius_km

Step 4	Admin Contact Setup
	Designate a primary contact person and configure additional admin users.
	• Fields: contact_name, contact_phone, contact_email • Multiple admin users can be added post-onboarding

Step 5	Payment Method
	Add a billing card or bank account from which shift costs will be collected.
	• Card: saved via payment gateway token (no raw card stored) • Billing triggered automatically on timesheet approval

7. Data Model Schema

All tables use UUID primary keys. Timestamps are stored as UTC. The schema is designed for PostgreSQL with JSONB for flexible metadata fields.

Core identity table for all roles.

Users	Type	Note
id	UUID PK	Primary key
phone	VARCHAR(20) UNIQUE	Phone number
email	VARCHAR(255) UNIQUE	Email address
role	ENUM('worker', 'employer', 'admin')	User role
status	ENUM('pending', 'active', 'suspended', 'banned')	Account status
otp_secret	VARCHAR	Current OTP value
otp_expires	TIMESTAMP	OTP expiry
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

Extended profile data for worker-role users.

WorkerProfiles	Type	Note
id	UUID PK	
user_id	UUID FK → Users	
full_name	VARCHAR(100)	
dob	DATE	Date of birth
avatar_url	TEXT	Profile photo URL
bio	TEXT(500)	
location_lat / lng	DECIMAL(9,6)	Worker base location
radius_km	INT	Max work radius
kyc_status	ENUM('unverified', 'pending', 'approved', 'rejected')	
rating_avg	DECIMAL(3,2)	Computed rating
total_shifts	INT	Completed shift count

Education, experience, and certification records.

WorkerQualifications	Type	Note
id	UUID PK	
worker_id	UUID FK → WorkerProfiles	
type	ENUM('education','certification','work_experience')	
title	VARCHAR(200)	Degree / role / cert title
institution	VARCHAR(200)	School / company / issuer
from_date / to_date	DATE	
description	TEXT	
verified	BOOLEAN	Admin-verified flag

KYC documents uploaded by workers.

WorkerDocuments	Type	Note
id	UUID PK	
worker_id	UUID FK → WorkerProfiles	
type	ENUM('govt_id','right_to_work','background_check')	
file_url	TEXT	Secure storage URL
status	ENUM('pending','approved','rejected')	
reviewed_by	UUID FK → Users	Admin reviewer
reviewed_at	TIMESTAMP	

Work category taxonomy.

Categories	Type	Note
id	UUID PK	
name	VARCHAR(100)	e.g. Hospitality, Logistics
slug	VARCHAR(100) UNIQUE	URL-safe identifier
icon_url	TEXT	

Junction: workers ↔ categories.

WorkerCategories	Type	Note
worker_id	UUID FK	Composite PK
category_id	UUID FK	Composite PK

Extended profile for employer-role users.

EmployerProfiles	Type	Note
id	UUID PK	
user_id	UUID FK → Users	
company_name	VARCHAR(200)	
reg_number	VARCHAR(100)	Business registration
logo_url	TEXT	
address	TEXT	
city	VARCHAR(100)	
verified	BOOLEAN	Admin-verified company
rating_avg	DECIMAL(3,2)	

Individual shift postings by employers.

Shifts	Type	Note
id	UUID PK	
employer_id	UUID FK → EmployerProfiles	
category_id	UUID FK → Categories	
title	VARCHAR(200)	
description	TEXT	
location_lat / lng	DECIMAL(9,6)	Shift venue coordinates
address	TEXT	
date	DATE	
start_time / end_time	TIME	
hourly_rate	DECIMAL(8,2)	
currency	VARCHAR(5)	e.g. INR, USD
slots_total / slots_filled	INT	Headcount tracking
status	ENUM('draft', 'open', 'filled', 'cancelled', 'completed')	

Worker applications to shifts.

ShiftApplications	Type	Note
id	UUID PK	
shift_id	UUID FK → Shifts	
worker_id	UUID FK → WorkerProfiles	
status	ENUM('applied', 'shortlisted', 'confirmed', 'rejected', 'withdrawn')	
applied_at	TIMESTAMP	
updated_at	TIMESTAMP	

Clock-in/out records for each confirmed application.

Timesheets	Type	Note
id	UUID PK	
application_id	UUID FK → ShiftApplications	
clock_in / clock_out	TIMESTAMP	
clock_in_lat / lng	DECIMAL(9,6)	Geolocation at clock-in
total_hours	DECIMAL(5,2)	Calculated worked hours
status	ENUM('pending','approved','disputed','paid')	
approved_by	UUID FK → Users	
approved_at	TIMESTAMP	

Financial record per approved timesheet.

Payments	Type	Note
id	UUID PK	
timesheet_id	UUID FK → Timesheets	
worker_id / employer_id	UUID FK	
gross_amount	DECIMAL(10,2)	Amount charged to employer
platform_fee	DECIMAL(10,2)	Platform commission
net_amount	DECIMAL(10,2)	Amount paid to worker
currency	VARCHAR(5)	
status	ENUM('pending','processing','completed','failed')	
gateway_ref	VARCHAR(200)	Payment gateway transaction ID
paid_at	TIMESTAMP	

Mutual ratings after shift completion.

Ratings	Type	Note
id	UUID PK	
shift_id	UUID FK	
reviewer_id / reviewee_id	UUID FK → Users	
reviewer_role	ENUM('worker','employer')	
score	INT (1-5)	
comment	TEXT	
created_at	TIMESTAMP	

In-app and push notification records.

Notifications	Type	Note
id	UUID PK	
user_id	UUID FK	
type	VARCHAR(100)	e.g. shift_confirmed, payment_sent
title / body	VARCHAR / TEXT	
read	BOOLEAN	
meta	JSONB	Additional payload data
created_at	TIMESTAMP	

8. REST API Structure

All endpoints are versioned under `/api/v1/`. Authentication uses Bearer JWT tokens in the Authorization header. Responses follow a consistent envelope: `{ "status": "success"|"error", "data": {}, "message": "" }`.

Authentication

POST	<code>/api/v1/auth/request-otp</code>	Request OTP via phone or email
POST	<code>/api/v1/auth/verify-otp</code>	Verify OTP and receive JWT pair
POST	<code>/api/v1/auth/refresh-token</code>	Refresh access token
POST	<code>/api/v1/auth/logout</code>	Invalidate refresh token

Worker — Profile & Onboarding

GET	<code>/api/v1/workers/me</code>	Get own profile
POST	<code>/api/v1/workers/profile</code>	Create worker profile (step 2–3)
PUT	<code>/api/v1/workers/profile</code>	Update worker profile
PUT	<code>/api/v1/workers/categories</code>	Set categories of interest
POST	<code>/api/v1/workers/qualifications</code>	Add qualification record
DELETE	<code>/api/v1/workers/qualifications/:id</code>	Remove qualification
POST	<code>/api/v1/workers/documents</code>	Upload KYC document

Worker — Shifts

GET	<code>/api/v1/shifts</code>	List shifts (filters: category, date, lat, lng, radius, rate)
GET	<code>/api/v1/shifts/:id</code>	Get shift details
POST	<code>/api/v1/shifts/:id/apply</code>	Apply for a shift
DELETE	<code>/api/v1/shifts/:id/apply</code>	Withdraw application
GET	<code>/api/v1/workers/applications</code>	List own applications

Employer — Profile & Onboarding

GET	<code>/api/v1/employers/me</code>	Get own employer profile
POST	<code>/api/v1/employers/profile</code>	Create employer profile
PUT	<code>/api/v1/employers/profile</code>	Update employer profile

Employer — Shifts

POST	<code>/api/v1/employers/shifts</code>	Create a new shift
PUT	<code>/api/v1/employers/shifts/:id</code>	Edit shift (while open)

DELETE	/api/v1/employers/shifts/:id	Cancel shift
GET	/api/v1/employers/shifts	List own shifts
GET	/api/v1/employers/shifts/:id/applicants	List shift applicants
PUT	/api/v1/employers/shifts/:id/applicants/:appId	Update applicant status

Timesheets

POST	/api/v1/timesheets/:applicationId/clock-in	Worker clocks in (geolocation required)
POST	/api/v1/timesheets/:applicationId/clock-out	Worker clocks out
GET	/api/v1/timesheets/:id	Get timesheet detail
PUT	/api/v1/timesheets/:id/approve	Employer approves timesheet
POST	/api/v1/timesheets/:id/dispute	Raise timesheet dispute

Payments

GET	/api/v1/payments/worker/history	Worker earnings history
GET	/api/v1/payments/employer/history	Employer billing history

Ratings

POST	/api/v1/ratings	Submit a rating (worker or employer)
GET	/api/v1/ratings/worker/:workerId	Get worker ratings
GET	/api/v1/ratings/employer/:employerId	Get employer ratings

Notifications

GET	/api/v1/notifications	List notifications (paginated)
PUT	/api/v1/notifications/:id/read	Mark as read
PUT	/api/v1/notifications/read-all	Mark all as read

9. Shift Lifecycle

The shift lifecycle represents the end-to-end journey of a single shift from creation through to payment and rating. Each state transition triggers notifications to relevant parties.

Stage	Actor	Status Change	Description
Shift Created	Employer	DRAFT → OPEN	Employer fills shift form. On publish, shift status moves to OPEN and becomes discoverable by matching workers.
Worker Applies	Worker	APPLICATION: applied	Worker taps Apply. ShiftApplication record created with status "applied". Employer notified.
Employer Reviews	Employer	APPLICATION: shortlisted / rejected	Employer views applicant list, reviews profiles and ratings. Can shortlist candidates or reject.
Shift Confirmed	Employer	APPLICATION: confirmed / SHIFT: filled	Employer accepts required number of workers. Workers receive push notification. Shift status moves to FILLED when all slots are taken.
Shift Day — Clock In	Worker	TIMESHEET: created	Worker must be within the defined geofence radius (default 200m) to clock in. GPS coordinates logged. Employer notified of clock-in.
Shift Day — Clock Out	Worker	TIMESHEET: pending	Worker clocks out at end of shift. Total hours calculated. Timesheet moves to PENDING APPROVAL.
Timesheet Approved	Employer	TIMESHEET: approved	Employer reviews logged hours. Can approve as-is, edit hours (with note), or raise a dispute. On approval, payment is automatically triggered.
Payment Processed	Platform	PAYMENT: completed	Platform deducts gross amount from employer. Net amount (gross minus platform fee) is transferred to worker payout account. Payment record created.
Mutual Rating	Both	POST-SHIFT	Both worker and employer are prompted to rate each other (1–5 stars + optional comment). Ratings update both parties' cumulative averages.

End of Document — GigWork Project Specification v1.0